

Topology Preparation

Session 8

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*Unless otherwise stated, materials are taken from UvA's very own Dutch-language *Syllabus Algebra I* by Taelman et al. (2015) and my university's choice of an undergraduate abstract algebra textbook, the third edition of *Abstract Algebra* by Herstein (1996). UvA's book is good, I just think some things are missing... so we have a supplementary textbook.

4 More Examples: Finite Groups, Groups of Transformations

Getting used to groups so far? Good! Now let's discover some other groups that might not be that familiar.

Example 4.1. Remember $\mathbb{Z}_n$? It is the set of congruence classes $\mathbb{Z}_n = \{[0], [1], \dots, [n-1]\}$ given a fixed integer $n > 1$. Recall that two integers belong to the same congruence class if and only if they have the same remainder upon integer division by n .

Now when you add two integers, what happens to their remainders? Let $a_1, a_2 \in \mathbb{Z}$ each be divided by n so that $a_1 = q_1n + r_1$ and $a_2 = q_2n + r_2$ for some $q_1, r_1, q_2, r_2 \in \mathbb{Z}$ with $0 \leq r_1 < n$ and $0 \leq r_2 < n$. Then

$$a_1 + a_2 = q_1n + r_1 + q_2n + r_2 = (q_1 + q_2)n + (r_1 + r_2).$$

Loosely speaking, the remainder of the sum of two integers is the sum of the remainders. Well, kinda. If the sum of the remainder, that is, $r_1 + r_2$, ends up being greater than n , then of course the actual remainder of the sum is first divided through by n again, and only then we take that remainder.

So, why bother? Well, this observation means we can define

$$[a] + [b] = [a + b]$$

for $a, b \in \mathbb{Z}$. This is unambiguous because we consider their remainders, as above. This time, $a + b$ possibly being more than n does not become a problem because it is the congruence class that we consider, not $a + b$ itself.

This allows $\mathbb{Z}_n$ to form an abelian group under this addition operation, called the *restklassengroep* in Dutch, literally "group of remainder classes". Closure is clear. (G1) follows from the associativity of $+$ in $\mathbb{Z}$: for all $a, b, c \in \mathbb{Z}$,

$$\begin{aligned} ([a] + [b]) + [c] &= [(a + b)] + [c] \\ &= [(a + b) + c] = [a + (b + c)] \\ &= [a] + [(b + c)] = [a] + ([b] + [c]). \end{aligned}$$

For (G2), the identity element is $[0]$, which should be apparent just from how this addition operation is defined. Also clear is (G3), with $-[a] = [-a]$, and (G4) follows from the commutativity of $+$ in $\mathbb{Z}$.

Remark. Unambiguity or well-defined-ness of an operation is best shown by proving that for any pair of inputs, if they are equal then the outputs are equal, so to speak. For the addition operation in $\mathbb{Z}_n$ as above, considering that the inputs are $[a]$ and $[b]$, if $[a_1] = [a_2]$ and $[b_1] = [b_2]$, we prove that $[a_1 + b_1] = [a_2 + b_2]$. Since a_1 and a_2 are in the same congruence class, it must be that $n \mid a_1 - a_2$, and similarly $n \mid b_1 - b_2$, so $n \mid (a_1 - a_2) + (b_1 - b_2)$ and thus $n \mid (a_1 + b_1) - (a_2 + b_2)$. Then by definition, $(a_1 + b_1)$ and $(a_2 + b_2)$ are in the same class, and so $[a_1 + b_1] = [a_2 + b_2]$.

References

Herstein, I. N. (1996). *Abstract Algebra* (3 ed.). Prentice-Hall.

Taelman, L., Lenstra, H. W., & Oort, F. (2015). *Syllabus Algebra I*. <https://web.archive.org/web/20231121002503/https://staff.fnwi.uva.nl/l.d.j.taelman/algebra1.pdf>